Chapter 8
Strategic Visual Representation

Quotes

"…but I'll know it when I see it…"
Potter Stewart, Supreme Court Justice

This was a portion of a quote that related to Justice Potter Stewart referring to pornography. In a similar vein, although there are some very specific "science" around creating effective visuals, there is still a component of art involved.

"There are three kinds of lies: lies, damned lies and statistics"
Benjamin Disraeli, British Prime Minister

A cautionary note that you as a data scientist wield a tremendous amount of power in how you convey information.

Visualizations

There's a big difference between presenting facts and conveying meaning. Visualization enhances your ability to communicate information to your audience

Always look for ways to make fewer words from the existing words and also always look for a way to visually represent the words through a table or graphic. In general, follow these guidelines:

- Represent information clearly, concisely and coherently
- Help your audience see the information and discern relationships
- Don't misrepresent or "lie"
- Focus on the needs of the audience

Just because a visualization is beautiful doesn't mean it's effective. Don't just create your slides, design them!

Audience Drivers

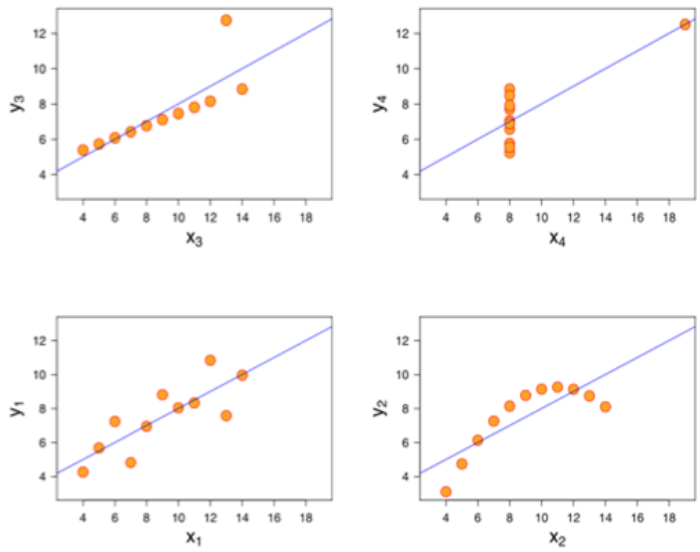
Take an audience perspective to improve your slides.

- What information does the audience need?
- How much detail does the audience need?
- What are desired actions / decisions?
- What visual would be easiest to understand?
- Does the culture impact the visualizations?

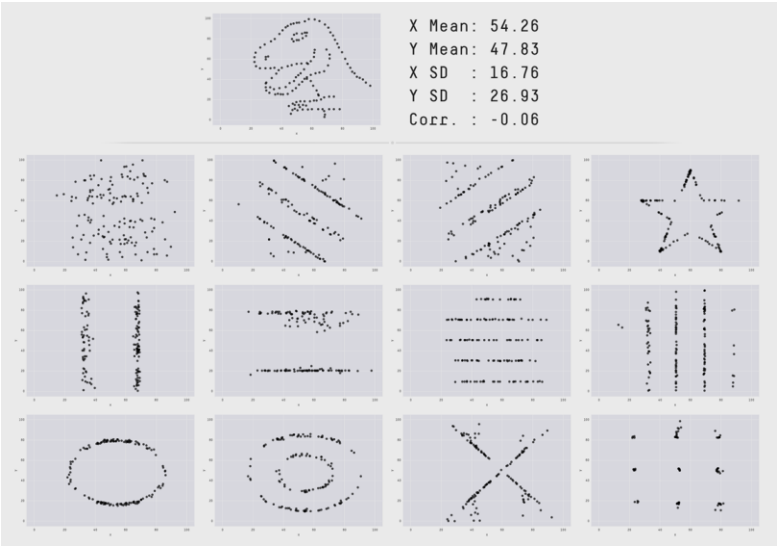
Visuals over Words

Although we have ways to specifically describe information with metrics. Those metrics don't always allow us to "see" the information. Make sure to look at the data.

Property	Value
Mean of x in each case	9 (exact)
Variance of x in each case	11 (exact)
Mean of y in each case	7.50 (to 2 decimal places)
Variance of y in each case	4.122 or 4.127 (to 3 decimal places)
Correlation between x and y in each case	0.816 (to 3 decimal places)
Linear regression line in each case	$y = 3.00 + 0.500x$ (to 2 and 3 decimal places, respectively)



Another example of the descriptors not truly showing the visual.



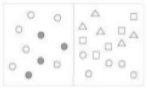
Source: https://www.autodeskresearch.com/publications/samestats

Implied Meaning of Visuals

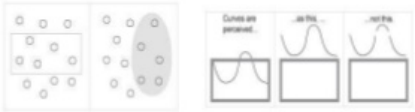
Proximity: Objects that are close together or connected are perceived as a group.



Similarity: Objects that share similar attributes (e.g. color, shape, size, pattern) are perceived as a group.



Enclosure and Continuity: Objects that appear to have a boundary or a continuation around them are perceived as a group.



Closure: Open structure can be easily perceived as closed and / or complete



There are a broad range of visual options available including size, shape, brightness, color, orientation and spacing. , take advantage of the multitude of visual representations:

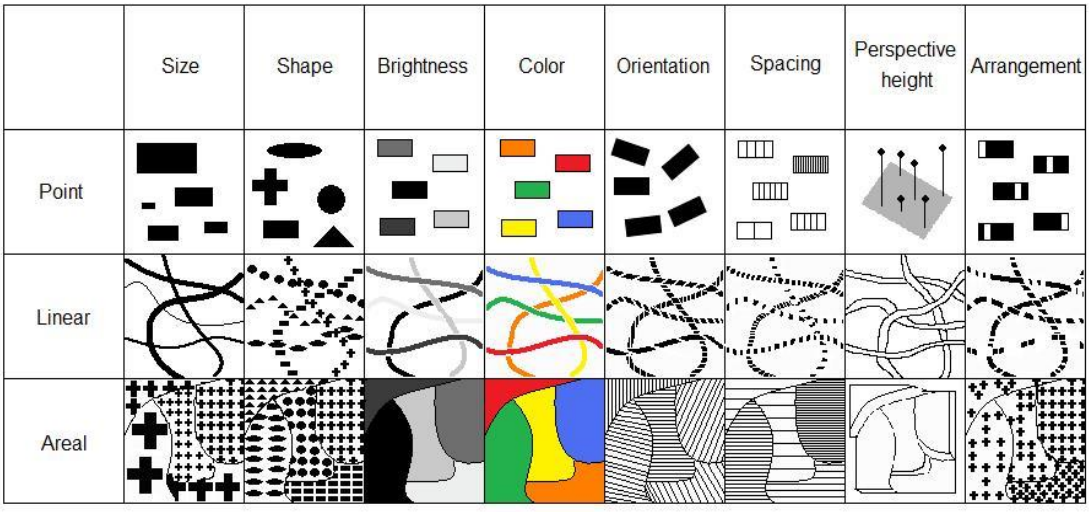
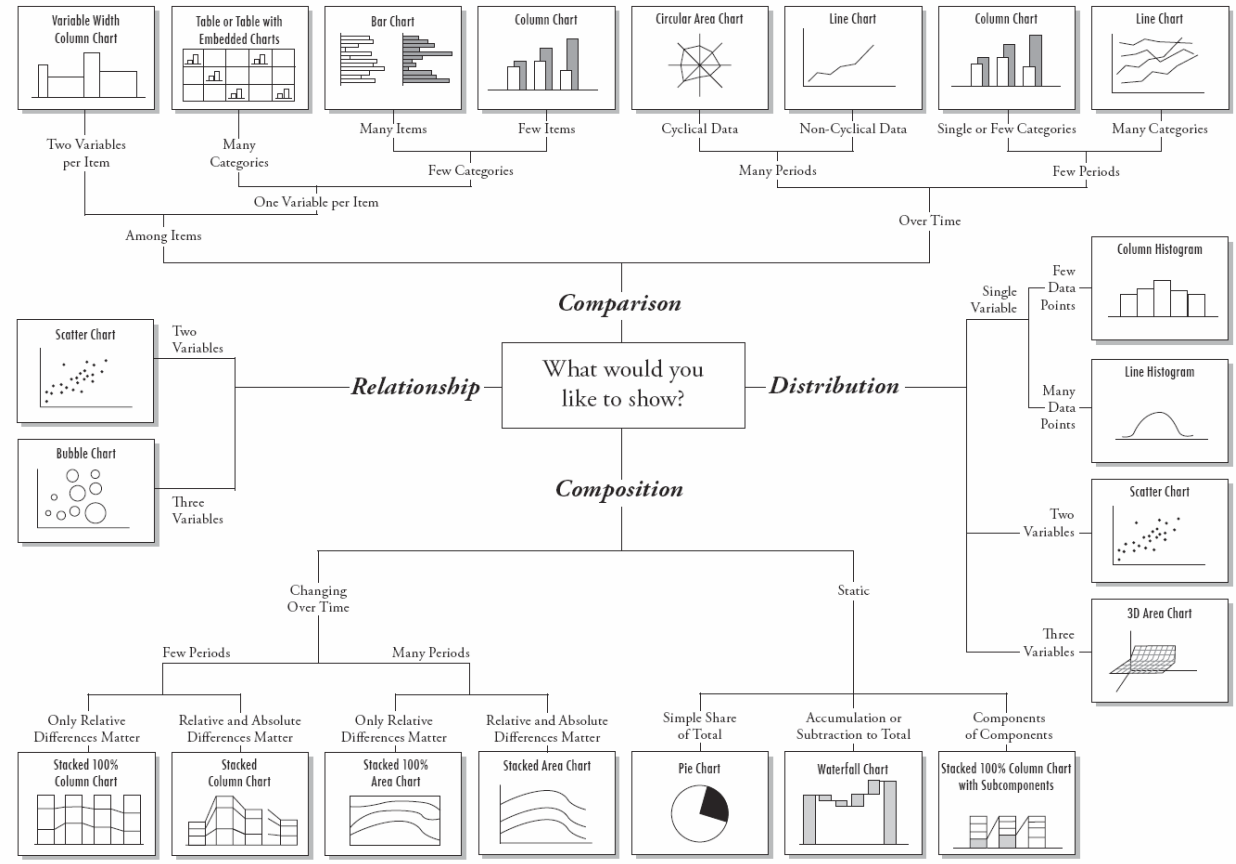


Chart Selection

Understanding the right way to display information is important. This is especially true when display information in the form of a chart. Although not an exact science, there is a reasonable way to select the chart type addresses the data and convey the information appropriately.

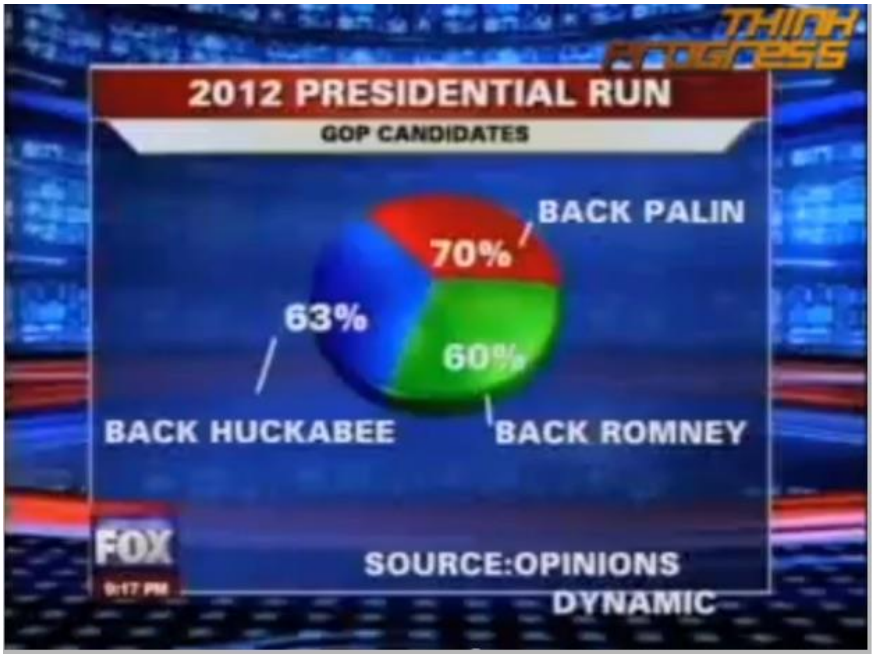


Source: http://www.labnol.org/software/find-right-chart-type-for-your-data/6523/

Confusing Charts

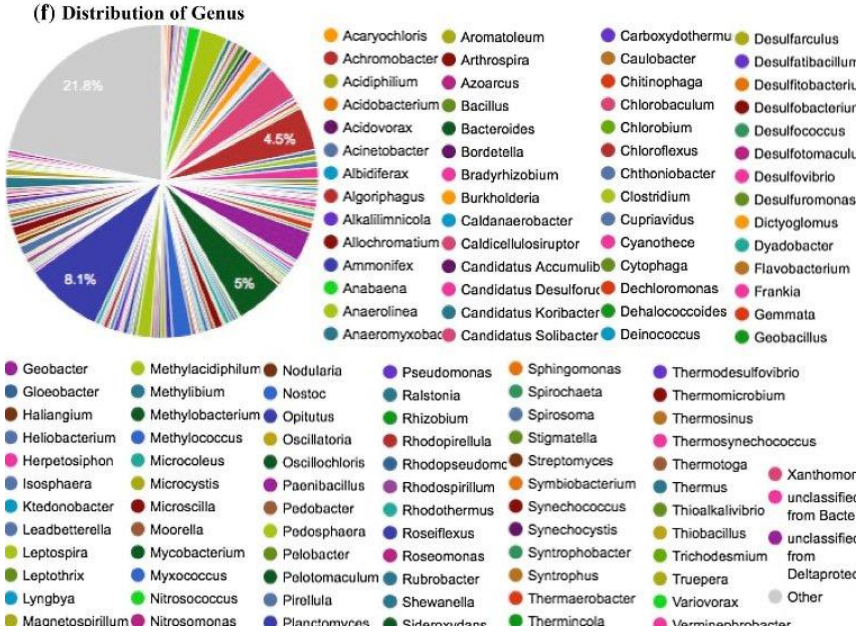
Make sure you convey the information correctly. There are real world examples where the visual does not help the audience.

Here's a chart that is confusing, at best. How can a pie add up to over 100%?

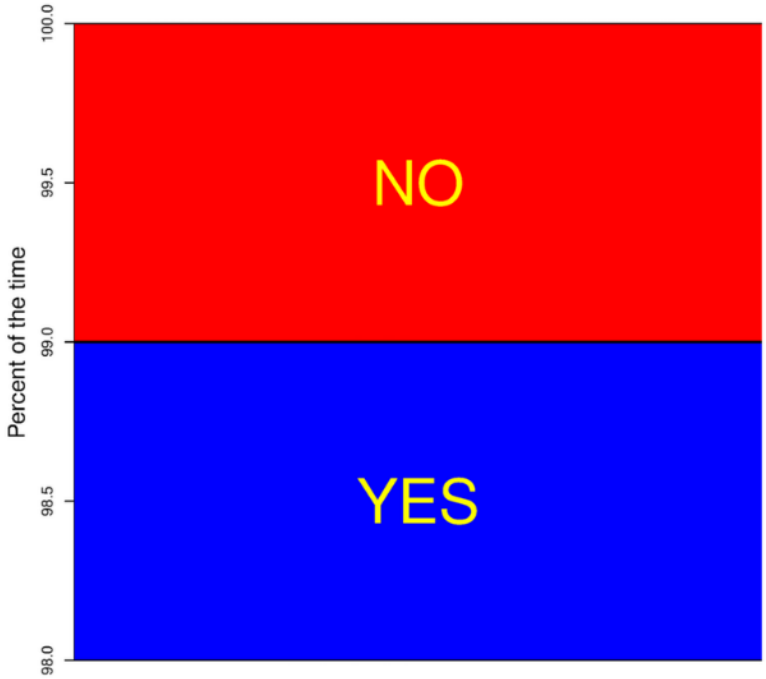


Source: Fox News

Here's a chart where it's impossible to understand the information being displayed, this is unfathomable at best. How can you match the Genus to the distribution percent?



Here's a chart that is misleading unless you are careful in reading the axis carefully – are the responses balance between yes and no?



Slide Best Practices

Develop "Best Practices" when communicating with slides.

- Don't be sloppy! Design each slide carefully
- Focus on what's important
    - Consolidate and simplify
    - Eliminate distractions
- Always use a legend…don't assume the audience will understand your color selection. Even if you use red, yellow, and green still use a legend to make it clear.
- Use highlighting or call outs to emphasize key information. Use this especially if there are lots of words or numbers. For example, a heat map over a large table of numbers makes it much easier to understand. Or, highlight a few key words in a paragraph to convey the overall meaning without losing the detail
- Get rid of text--visualize the story! Think: "Enough, but no more!" You need to provide enough information for the deck to stand on its own, but you don't want to have any more information than that. This is a balancing act and requires effort to find the right combination.

If you look at it and it doesn't make immediate sense, then there's something wrong! Even better, ask someone else to look at the chart. If they can't tell you the meaning quickly, you need to work on making it easier to understand.

Visualization Challenges

Also, be on the lookout for typical visualization problems.

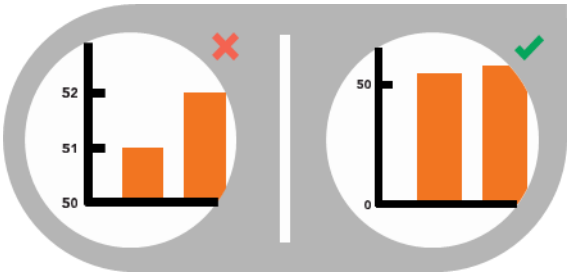
- Inappropriate display choice. For example, pick the right chart to convey the information.
- Variety for the sake of variety. Don't have multiple different color boxes on a slide if each box doesn't have special meaning – keep them all the same color. If there is special meaning, then there should be a legend with each color and a description of the meaning.
- Too much information – visual clutter. Present just enough information, don't try to overdo it. For example, in general 3D charts don't provide any additional information, they're just clutter
- Abuse of color. Don't go with extreme colors. For example, use of saturated, bright, or inappropriate colors doesn't help the audience. Also, don't completely rely on color to convey meaning…remember audience members may be color blind. For example, in displaying a stop light (red, yellow, green) dashboard, consider putting the letters "R", "Y", and "G" in the stoplights to make it clear.
- Poor choices for item display
    - Noisy fill patterns.
    - Noisy / heavy line styles
- Inconsistent ordering and placement
- Inconsistent or reversed scales
- Bad use of data
    - encoding data inaccurately
    - using counts vs. percentages
- Repeating words or patterns -- suggests reorganization. If you notice that you're repeating words or patterns on a slide or across a few slides, consider working on the slide and finding a way to display the information without repeating the words or patterns.

Chart Best Practices

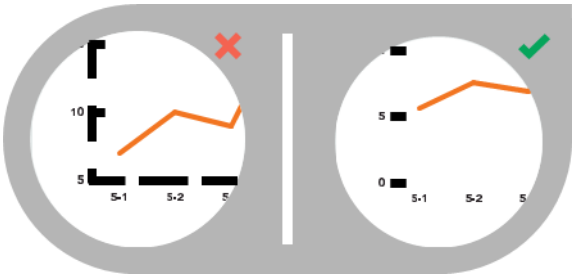
Note: the graphics for this section are from “Hubspot Data Visualization 101”.

Start the Y-Axis at Zero.

Cutting the Y-Axis is one of the most controversial practices in data visualization. Starting at a value above zero truncates the bars and doesn’t accurately reflect the full value. This, in some cases this can even be misleading.



Although a line chart does not have to start at a zero baseline, it should be included if possible. If relatively small fluctuations in data are meaningful (e.g., in stock market data), you may truncate the scale to showcase these variances.



Starting at zero also applies for area charts and bubble charts.

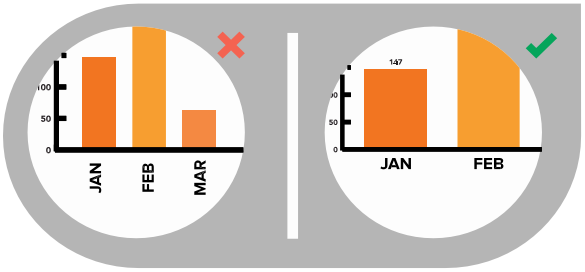
Whenever the axis does not start from zero, it is typically indicated on the axis with breaks in the line. This gives the viewer a visual cue that the axis is not starting from zero as might be expected



Bar Charts.

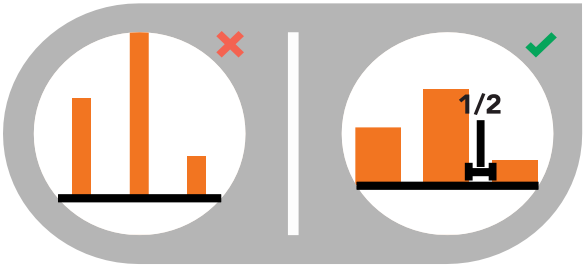
Use horizontal labels

Avoid steep diagonal or vertical type, as it can be difficult to read.



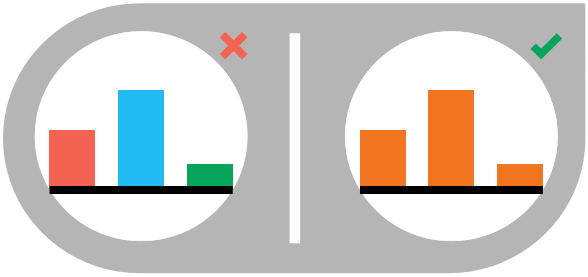
Space bars appropriately

Space between bars should be ½ bar width



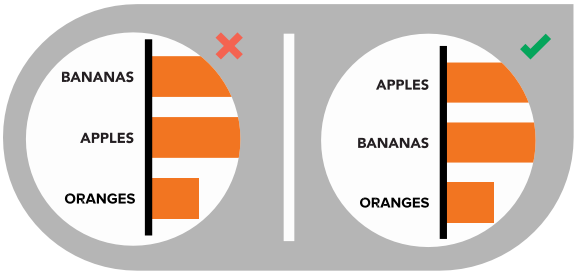
Use consistent colors

Use one color for bar charts. You may use an accent color to highlight a significant data point



Order data appropriately

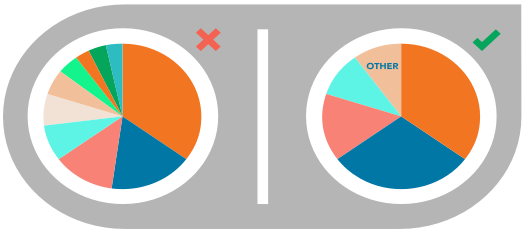
Order categories alphabetically, sequentially or by value. Ordering by value can make the graph much more insightful



Pie Charts

Visualize no more than 5 categories per chart.

It is difficult to differentiate between small values; too many slices decreases the impact of the visualization. If needed, group smaller values into an “other” or “miscellaneous” category, but make sure it does not hide interesting or significant information.



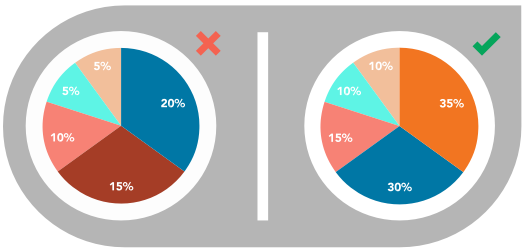
Don’t use multiple pie charts for comparison

Slice sizes are very difficult to compare side-by-side. Use a stacked bar chart instead.



Make sure data adds up to 100%

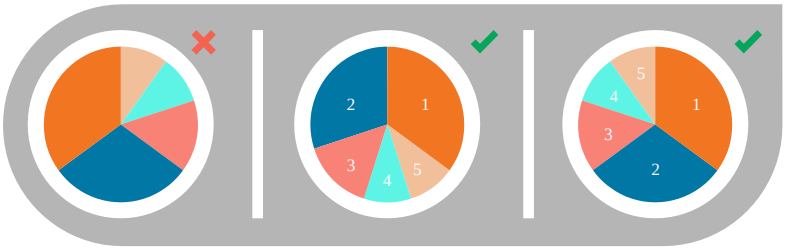
Verify that values total 100% and that pie slices are sized proportionate to their corresponding value.



Order slices correctly

There are two ways to order sections to aid comprehension:

- Place largest section at 12:00, going clockwise. Place 2nd largest section at 12:00 going counterclockwise. The remaining sections can be placed below, continuing counterclockwise.
- Start the largest section at 12 o’clock. Place remaining sections in descending order, going clockwise.



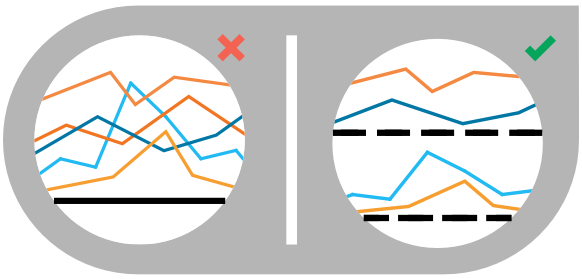
Line Charts

Displaying slope

When you're trying to demonstrate slope, make sure scales for X and Y are set accordingly. In other words, a slop of 45 degrees should look like 45 degrees to the viewer.

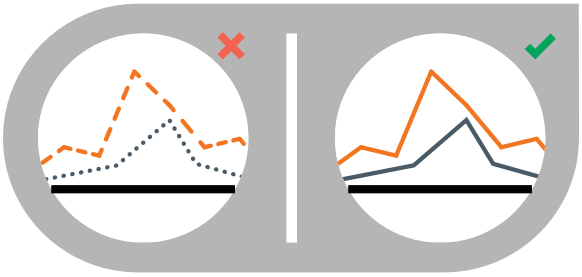
Don't Plot more than 4 lines

A line chart with too many lines is often referred to as a spaghetti graph. If you need to display more than 4 lines, break them out into separate charts for better comparison.



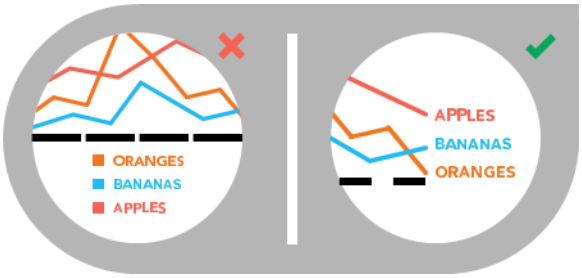
Use solid lines only

Dashed and dotted lines can be distracting.



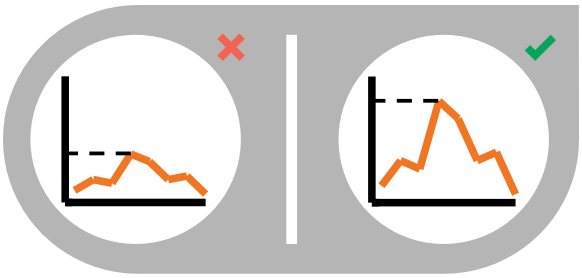
Label the lines directly

This lets readers quickly identify lines and corresponding labels instead of referencing a legend.



Use the right height

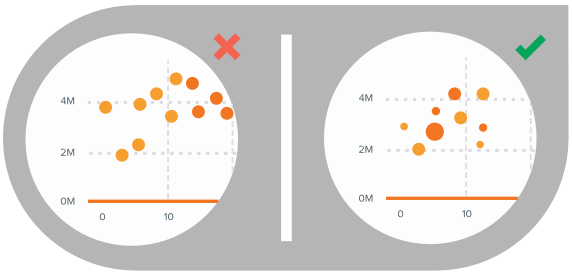
Plot all data points so that the line chart takes up approximately two-thirds of the y-axis' total scale.



Scatter Plots

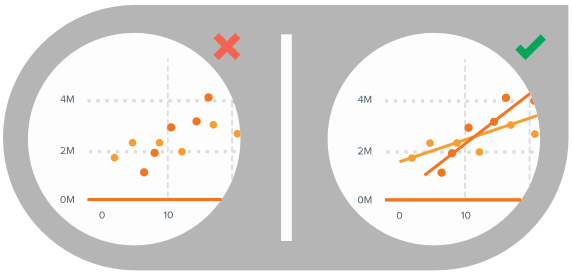
Include more variables

Use size and dot color to encode additional data variables.



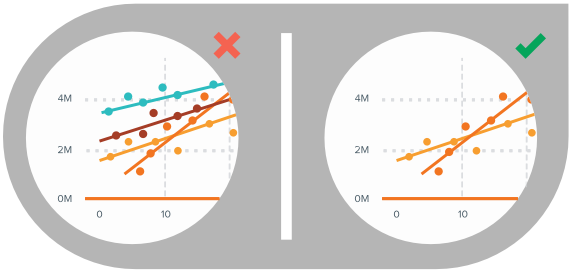
Use Trend lines

These help draw correlation between the variables to show trends.



Don’t compare more than 2 trend lines

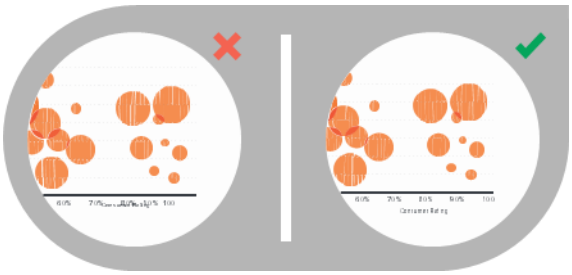
Too many lines make data difficult to interpret.



Bubble Charts

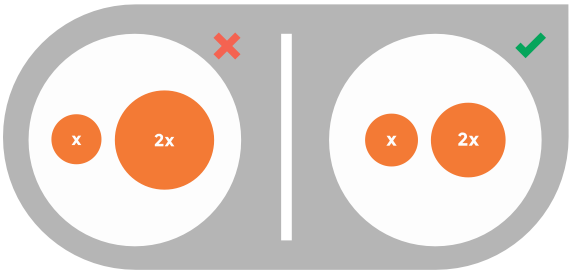
Make sure labels are visible

All labels should be unobstructed and easily identified with the corresponding bubble.



Size bubbles appropriately

Bubbles should be scaled according to area, not diameter.



Don’t use odd shapes

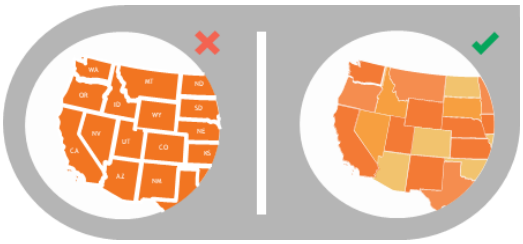
Avoid adding too much detail or using shapes that are not entirely circular; this can lead to inaccuracies.



Heat Maps

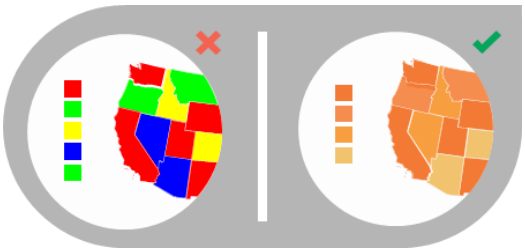
Use a simple map outline

These lines are meant to frame the data, not distract.



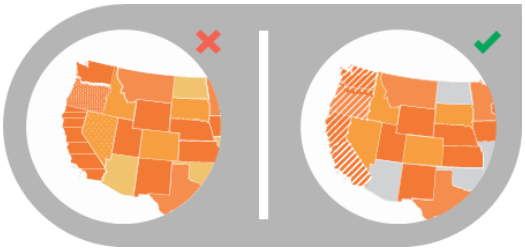
Select colors appropriately

Some colors stand out more than others, giving unnecessary weight to that data. Instead, use a single color with varying shade or a spectrum between two analogous colors to show intensity. Also remember to intuitively code color intensity according to values.



Use patterns sparingly

A pattern overlay that indicates a second variable is acceptable, but using multiple is overwhelming.



Choose appropriate data ranges

Select 3-5 numerical ranges that enable fairly even distribution of data between them. Use +/- signs to extend high and low ranges.



Optional Resources:

The following websites provide good information about what is good (or bad) about visualization.

- Edward Tufte – http://www.edwardtufte.com
- Data to Viz – http://www.data-to-viz.com (more detailed chart selection and also view the "caveats")
- Hubspot Data Visualization 101 - https://blog.hubspot.com/marketing/types-of-graphs-for-data-visualization
- Kaiser Fung – http://junkcharts.typepad.com
- Jen Underwood – http://www.jenunderwood.com
- Stephen Few – http://www.perceptualedge.com
- Flowing Data – http://www.flowingdata.com
- Infosthetics – http://www.infosthetics.com
- Visual.ly – http://www.visual.ly
- KDnuggets Data Visualization Cheat Sheet https://www.kdnuggets.com/2018/08/data-visualization-cheatsheet.html